

ARCHITECTURE OF PROCESS MINING BASED BUSINESS PROCESS OPTIMIZATION

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Abstract

The optimization of processes of business guarantees the successes of informationization project. Although its importance is fully recognized, it never obtains sufficient study. This paper first discuss the current business process optimization methods, they are classified into three type: participation-type, principle-type and analytical-type. Then, architecture of process mining based business process optimization is proposed. According to this architecture, the subset of information that is relevant for process mining or analysis is extracted from a variety of business process system. This information is then modelled in certain of expression. Based on the model, hidden relationship and unknown knowledge about business processes are extracted and stored in the form of business knowledge model. With the help of the acquired business knowledge, business process can be optimized both on macro structure level through case based reasoning and on micro parameter level through process simulation and linear programming. Business process performance measurement system is employed in the architecture to check effect of optimization. Finally, key issues in this architecture are discussed.

1 Introduction

The optimization of an organization's business processes is the core of all the questions about the organization performance improvement. It is also the key issue of business process reengineering, business process management (BPM), business process redesign, and so on. A useful working definition of BPM is given^[4] as systematic analyses, optimization, control, measurement, and management of all company processes to improve the company's performance. BPM helps in optimization a process in order to enhance its performance. Academics and business practitioners have been developing methodologies to support the application of BPM principles. However, most methodologies generally lack actual guidance on deriving a process design/optimization, thereby threatening the success of BPM^[14]. Indeed, a survey has proved that 85% of projects fail or experience problems^[16]. Process mining (namely workflow mining earlier) is a topic that has recently attracted the attention of various researchers.

According to W.M.P. van der Aalst, "the goal of process mining is to extract an explicit process model from event logs, i.e., the challenge to create a process model given a log with events such that the model is consistent with the observed dynamic behavior^[20]", "is to reverse the process and collect data at runtime to support workflow design and analysis^[19]". Process mining tools can answer business questions that traditionally were too time-consuming to resolve. They scour history data about the process operation for hidden patterns, finding predictive information that experts may miss because it lies outside their expectation.

The new, formerly hidden knowledge generated by Process mining tools can serve as a basis for process managers to optimize business processes so as to suit current business development and challenges and to remain at a competitive level with other business organizations.

This article presents architecture for business process optimization (BPO) based on process mining. In this architecture, the process mining approaches extracts business process model representations from history data and event logs in business information systems such as Workflow Management System or ERP systems, thus enables broader data mining analyses of apparent process behavior. Business process performance measurement model and analyses model are employed as a part of the architecture.

The paper is organized as follows. Section 2 reviews principles of business process optimization. Section 3 introduces the concept of process mining. Section 4 proposes architecture of business process optimization based on process mining and discuss the key issues in this architecture. Section 5 concludes the paper.

2 Principles of business process optimization

The efficiency of business processes is vital to the success of the company. Companies need to make sure that they carry out their business processes in the most efficient way. They need to design the processes to optimize their customer service and to streamline their coordination with vendors and other external partners.

The business process optimization methods are able to be classified to three types: participation-type, principle-type and analytical-type.

Participation-type business process optimization employs a team consists of process specialists, process operators and process managers, etc. Business process alternatives are discussed and evaluated by the team. Interviews and

workshops are organized to elicit important aspects of the business processes. This is, however, a tedious process and suffers from subjective, fragmented, and possibly unreliable sources of data. Both Peppard^[7] and Sharp^[1] discuss this type of method in detail.

Principle-type business process optimization, strictly speaking, is not really a type of optimization method. It usually consists of several pieces of technical or heuristic principles that guide the design of business processes. For example, Hammer^[12] in his seminal paper proposes seven principles of reengineering. Rupp and Russell^[13] summarize three golden rules and sixteen simple commandments for process design. Huang^[10] concludes eight “optimum rule of business process” from three perspectives of management, information flow and IT. Li^[11] argues five kinds of tactics to optimize the existing processes, such as choosing key process tactic, value increment tactic, integration tactic, concurrent tactic and out sourcing tactic.

Analytical-type business process optimization aims at using formal theory and techniques to model and derive the process design. It mainly focuses on the identification of processes parameters, and gains the desired solution through certain of algorithm and/or logic reasoning. For example, Orman^[9] proposes a “model management approach” to business process optimization. In this approach, processes are viewed as collections of decision models. Reoptimization of business processes is formulated as a dynamic programming problem. Aldowaisan^[15] provides a structural definition of the redesign search space using the number of employee types and the number of activities, and propose a systematic approach that combines some linear programming tools with unique BPR principles to support the technical redesign of the current process. Focusing on the re-engineering of knock-out processes, W.M.P. van der Aalst^[18] concentrates on the order in which these tasks need to be executed to yield an optimal process with respect to the utilization of resources and flow time. Hofacker^[5] treats business process as a set of activities, and the optimization of business process as a procedure of selecting best subset from underlying activities set. Each activity uses one or more inputs and generates one or more outputs. Both inputs and outputs are resources. Process optimization seeks to get sequential activities subset in which each activity can get its required resources when it starts. The seeking process may use three techniques: mathematical programming, a branch and bound method, and genetic algorithms. Rajiv Dewan^[3] discusses how to shorten the workflow execution cycle time by combing tasks. Tian^[2] continues Rajiv Dewan’s work and finds that the final result is irrelevant with the order of task combing, the combing problem is decomposed into a series of critical task combing.

3 Process mining

The term process mining refers to methods for distilling a structured process description or extracting of process knowledge from a set of real executions, or more specifically, from log data related to past process and application

executions. Respective logs are usually provided by workflow management systems, but also by other process-oriented applications, like enterprise resource planning or supply chain management systems. Typically, all these systems log event-based data (e.g., related to the start or completion of task executions) together with additional context information (e.g., about actors).

The main objective of process mining is to effectively use automatically collected data in order to gain process knowledge from the logs^[17]. In particular, process mining extracts formal process models from execution logs. Process mining applications include features of three categories^[6]:

- **Model construction** — The challenge of this feature is to dynamically construct business process models that are representative of the business flows found in the event logs.
- **Statistical performance analysis** — This feature aims to extract the development of predefined statistical measures (key performance indicators).
- **Knowledge discovery** — Combining data in an event log with data from other sources (i.e., data warehouses), this feature category enables users to search for hidden patterns and relationships.

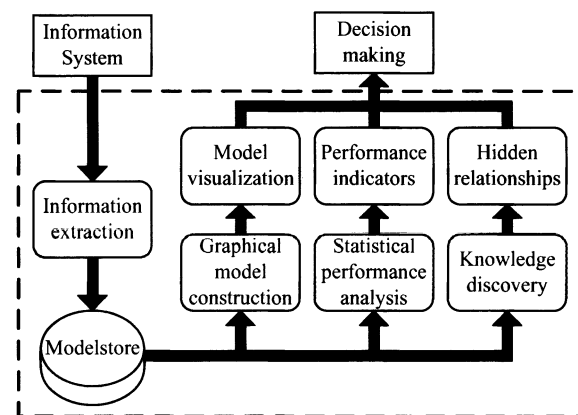


Figure 1: process mining application.

Figure 1 is an illustration of how process mining applications typically interact with decision makers and the information system.

4 Architecture of process mining based BPO

The establishment of architecture for business process optimization raises a number of challenges. This section outlines basic requirements (Section 4.1), gives an overview of the designed architecture (Section 4.2), and describes the key issues in the architecture (Section 4.3).

4.1 Requirements for the architecture

A process mining based BPO architecture must provide comprehensive facilities for business information extraction, representation, process knowledge discovery, for business process modification on both structure and parameter, for business process monitoring and must allow quick responses

to observed discrepancies between the optimized and the executed processes. In particular, process engineers should be supported in learning from previous (ad-hoc) instance changes and in deriving optimized process models or suggestion on process optimization from log data.

4.2 Overview of the architecture

Based on the requirements analysis in section 4.1, this paper proposes a process mining based BPO architecture as Figure 2.

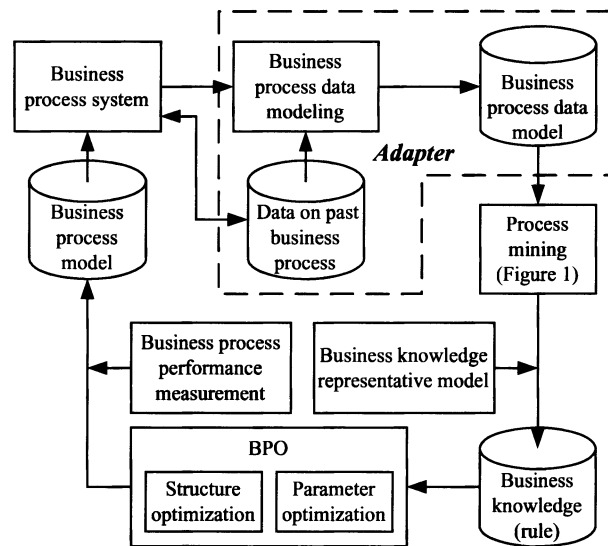


Figure 2: architecture of process mining based BPO.

Business tasks supported by the WfMS or ERP system or other business process system are logged as they are carried out by the business operators. Data on past business processes including vision, technology, management, sales, services, accountability and leadership is accumulated over time in databases. The subset of information that is relevant for process mining or analysis is then extracted from a variety of business process system. Because of the complexity and variance of the business process system, the information extraction can be performed using a specific adapter layer^[8]. This adapter layer contains knowledge of which event logs and attributes have to be accessed. The information extraction phase may also be responsible for dealing with missing values and misspellings. The information that has been extracted is then stored as records in a database in form of business process data model.

Based on this data source, the process mining application constructs graphical business process models and statistical performance indicators. Incorporating the data source with collections of other data sources, such as a data warehouse, provides a platform for mining for business knowledge, for example, unknown and valuable patterns and relationships. These patterns can be significant correlations between certain variables, such as specific departments, product categories, vendors, process paths, and process execution times.

Coupled with other information sources, such as interviews, surveys, on-the-job observations, work study, and so forth, the mined business knowledge is then able to be used to evaluate the business processes. If the process mining application provides dynamically constructed business process models, delta analyses^[19] are typically carried out to identify mismatches with models of best practice or TO-BE situations. These evaluations usually lead to a optimization of business processes, which again produces new inputs for the process mining application. For such reasons, process mining applications can be used as a tool for continuous monitoring and optimizing of business process.

4.3 Key issues in the architecture

Process mining based BPO architecture raises a number of issues. As indicated in the previous section, some of these issues have been answered while others require further research. Therefore, we review the key challenging issues.

(1) Business process data modelling

Data on past business processes is the source of knowledge gaining and business process optimization. The data and the information system in which the data exists vary in form of expression. This variance hinders business knowledge gaining through process mining. Therefore, modelling the data and build an expression method is the basic work in the architecture.

(2) Business knowledge representative model

The fruit of process mining is business knowledge that is able to be used to optimize business processes. But there is a promise that the business knowledge is well expressed. Therefore, we need construct a business knowledge expression model.

(3) Process mining method

Process optimization requires extract hidden, useful knowledge from huge amount of incomplete, noisily, fuzzy, and stochastic business data. Process mining method is the core issue in the BPO architecture.

(4) BPO based on business knowledge

Business process optimization includes macro structure optimization and micro parameter optimization. Both of them are closely connected with the business knowledge expression model. In macro structure optimization level, case based reasoning is a very useful problem solving approach. In micro parameter optimization level, process simulation, linear programming, etc are all available methods.

(5) Business process performance measurement

Process performance is the only criterion to measure the optimization effects. A business process performance measurement system is an indispensability part of the whole optimization architecture.

5 Conclusion

Optimized effective and efficient business processes are guarantee of the competitiveness of an enterprise. This paper discusses the current business process optimization methods and proposes architecture of process mining based business

process optimization. Finally, key issues in this architecture are discussed.

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